

EARLY DETECTION OF WILDFIRES WITH GOES-R TIME-SERIES AND DEEP GRU NETWORK

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ABSTRACT

In recent years, wildfires have become major devastating hazards that affect both public safety and the environment. Thus, agile detection of the wildfires is desirable to suppress wildfires in the early stage. Owing to the high temporal resolution, GOES-R satellites offer capabilities to obtain images every 15 minutes enabling a near real-time monitoring of wildfires. In this research, a time-series-based deep learning framework, composed of Gated Recurrent Units (GRU), is proposed to capture the emerging of the wildfire at early stage. By feeding the embedding of the coarse satellite imagery to Deep GRU network, the active fires are segmented out from the remote sensing imagery. The preliminary results show that proposed network can detect the wildfires earlier than the state-of-the-art fire product for 2020 wildfires in California and British Columbia, at the same time provide sufficiently high accuracy on the burned areas.

Index Terms— GOES-R, GRU, Deep Learning, Wildfire monitoring

1. INTRODUCTION

Wildfires have recently drawn increasing attention from the public, as the devastating wildfires caused huge human, economic and environmental losses. Fire monitoring based on satellite imagery is considered of great importance to detect active wildfire at the early stage.[1]. Thus, various wildfire products have been developed to provide near real-time fire information. For active fire monitoring, Visible Infrared Imaging Radiometer Suite (VIIRS)[2] and the Moderate Resolution Imaging Spectroradiometer (MODIS)[3][4] are often used. MODIS and VIIRS are able to acquire images twice a day. In particular, VIIRS offers an higher spatial resolution (375 m), which also leads to a more accurate detection of active fires and burned areas[2]. In [2], VIIRS produced around 20 percent omission error and only 3 percent commission error globally. However, revisit time of half a day is still inadequate in early fire detection, given the fact that some wildfires spread fast in the early stage. GOES-R satellites are able to obtain images every 15 minutes thus have

potential for early detection and near real-time monitoring of wildfires. In [5], a fire product is developed to utilize the high temporal resolution nature of GOES-R satellites. However, the accuracy of the fire product is rather poor due to the low spatial resolution of GOES-R sensor (0.5, 1, 2km at nadir), with false alarm rate around 60% to 80% for medium and low confidence fire pixels.

The objective of this research is to develop a deep-learning model to detect active fire from GOES-R time series aiming to overcome the limitation of the low spatial resolution improving the overall accuracy. We propose a time-series based model to detect and segment active fire pixels in GOES-R imagery. Compared with classical image segmentation network used in remote sensing (e.g. CNN, U-NET), there are two main novelties. First, the segmentation utilizes the temporal information based on deep GRU network, which is guaranteed by the high temporal resolution of GOES-R imagery. Secondly, the approach is designed to exploit GOES-R low spatial resolution imagery. Hence, classical segmentation network such as Convolutional Neural Network is not applicable to GOES-R imagery due to the coarse spatial resolution. Therefore, unlike other segmentation techniques, the purposed network will work at pixel-level and perform pixel level classification to all the pixels in the GOES-R imagery.

2. METHODOLOGY

The main idea of the presented methodology is to use a Deep GRU network to detect the active fire in GOES-R time series using the simultaneous acquisitions of VIIRS sensor as reference time-frames. Hereafter we present the different processing steps.

2.1. GOES-R Preprocessing and reference data

The preprocessing of GOES-R imagery is performed using the Google Earth Engine platform [6]. To detect fire pixels, Band 7 from GOES-R satellites is often used; with wavelength equals to $\lambda_7 = 3.80 - 4.00\mu m$, this thermal band is capable of detecting the brightness and heat from the ground. Band 14 with wavelength $\lambda_{14} = 10.8 - 11.6\mu m$ is used to reduce the illumination difference between day and night. The

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active fire index is calculated based on the following equation:

$$I_{active_fire} = \frac{Band_7 - Band_{14}}{Band_7 + Band_{14}} \quad (1)$$

However, wildfires often appear together with smoke and clouds. Hence, Band 15 with wavelength $\lambda_{15} = 11.8 - 12.8\mu m$ are used to remove the cloud. The cloud mask is create based on following thresholding on Band 15, where:

$$Cloud_mask = Band_{15} > 280 \quad (2)$$

The ground truth is generated using the FIRMS[7] dataset based on VIIRS active fire product, which provides 375 m spatial resolution. For each fire pixel, the location and the brightness temperature value are provided.

For each generated ground truth image, the images of GOES-R are selected for past 3 hours before the acquisition time of the ground truth. To further remove the interference, the max aggregated image over all the images is served as the input to the network.

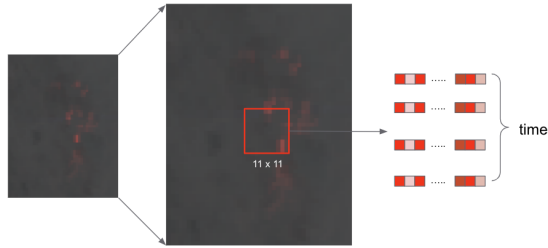


Fig. 1: Preprocessing process of the GOES-R imagery

Before the training stage, two main steps are required: 1) The first step is to linearly resample the image to the size of the ground truth to achieve the one-to-one mapping between the GOES-R images pixels and the ones of VIIRS reference data. Another one is that this resampling process will enhance the resolution, thus, provides higher accuracy in burned area mapping. 2) The second step is to patch surrounding pixels for each pixel in an 11 by 11 window and then flatten the image patch into an vector. By combining all vectors of different epochs, the time series of vectors are created as input to the network.

2.2. Deep GRU network

2.2.1. Gated Recurrent Unit

Gated Recurrent Unit is one type of Recurrent neural network proposed by [8]. Like classical recurrent neural network, for every output of the gated recurrent unit, it will combine the information from both the input at current timestamp and the output from the previous timestamp. Owing to this sequential architecture, recurrent neural network has become the standard to process time-series data. Because early fire detection

is a time-critical application, we choose GRU as the basic unit of the network to reduce the computation inference time.

2.2.2. Deep GRU network

In this section, we propose the Deep GRU Network to best utilize the time-series information for all the images. As shown in Figure 2, this network is implemented as a 6-level architecture, which consists of 5 GRU layers with many-to-many architecture and one level of Dense network to generate the output. After each GRU layer, hyperbolic tangent(tanh) is used as the activation function. There are two benefits of using many-to-many architecture: 1) one is that the network can generate outputs for all the timestamps at the same time, which reduces the redundancy when generating the input dataset, otherwise, there will be a sequence for all the timestamps. 2) another one is that this architecture make it possible to fully utilize the information between timestamps for each output at one timestamp.

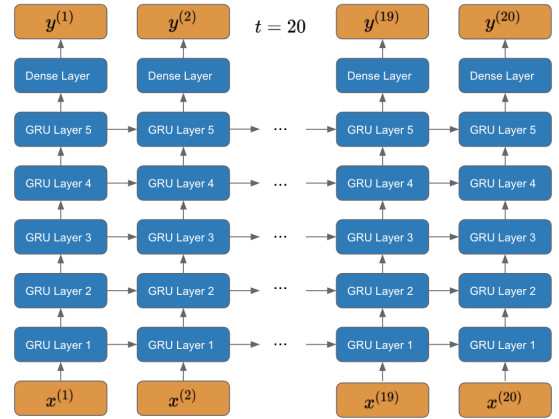


Fig. 2: Deep GRU network architecture

The network takes a sequence of 20 vectors of pixels as the input for 20 different timestamps. The sequence length of the GRU network is picked as 20 to fit the shortest length of the time-series of the wildfire in the training set. And for each vector, it corresponds to one pixel from the image, which will be discussed in the following section. For Level 1 and level 2, they increase the dimension of the vector to 256, which expand the features out of the input vector. Additionally, level 3, 4, 5 decrease the dimension of the vector to 32, which encodes the features into lower dimension representations. At the output layer, there is one single dense layer applied across all the timestamps, which generate the output for the corresponding pixel. As for the loss function, since the label is the brightness temperature value of the fire pixel, Deep GRU network uses MSE(Mean Squared Error) to minimize the difference between the label and the output.

2.3. Postprocessing

As mentioned in section 2.2, the network only generate a regression value and further postprocessing needs to binarize the result at the inference stage. From the Deep GRU network, we get the output value for each pixels in every time-stamps. By aliasing the pixels, the resulting image can be reconstructed. Then, we automatically threshold the image based on the mean value of the image to detect the anomaly pixels in the image and reduce the interference caused by the smoke and clouds.

3. STUDY AREAS AND DATA

Considering that the GOES-R satellites are geostationary and cover North America and South America regions, we selected the following areas for training and validating the proposed approach. The training dataset is composed of 4 historical wildfires from California in 2020: 1) August Complex Fire, 2) SCU Lighting Complex 3) CZU Lighting Complex 4) LNU Lighting Complex. For validation, two wildfires respectively

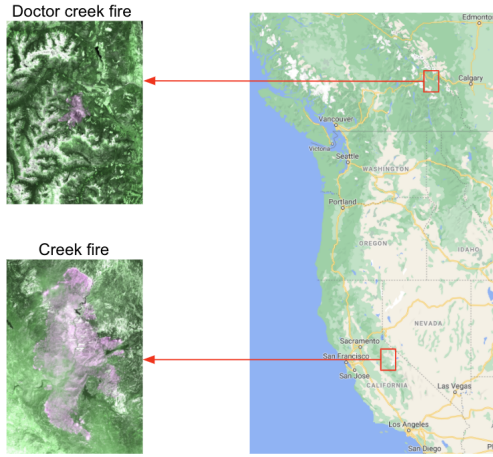


Fig. 3: Validation areas: Doctor Creek Fire (British Columbia, Canada), The Creek Fire (California, USA).

in California and in British Columbia were selected (see Fig. 3) : (1) Creek Fire, currently the fourth largest fire in California's history, started on September 4th, 2020. This wildfire caused 153,738 hectares of land burned which mostly belongs to Sierra National Forest. (2) The Doctor Creek Fire, started on September 19, is located at 25 kilometers south-west of Canal Flats, British Columbia, Canada. The size of this wildfire is 7,645 hectares. All the remote sensing training and validation images are collected from Google Earth Engine platform [6]. The training dataset comprises a total of 60000 samples of time series of pixel vectors. The network is trained with a minibatch size of 32 and the Adam optimizer with learning rate $l_r = 0.001$. The validation dataset is prepared for the two study areas and each image in the time series

comes from the max aggregation over all acquisitions within every hour.

4. RESULTS AND DISCUSSION

We performed the validation of the proposed method considering two key aspects: 1) earliest detection time 2) accuracy assessment on burned areas. For the earliest detection, the proposed method is compared with FIRMS VIIRS fire product, which provides active fire detection twice a day. The earliest detection time of the proposed method is based on earliest time when fire pixels are detected in the result images. For burned areas accuracy assessment, the Difference Normalized Burned Ratio(dNBR) is used to produce reference maps using high resolution (30m) pre- and post-fire Sentinel-2 images. The dNBR is calculate based on the following equation.

$$dNBR = NBR_{pre_fire} - NBR_{post_fire} \quad (3)$$

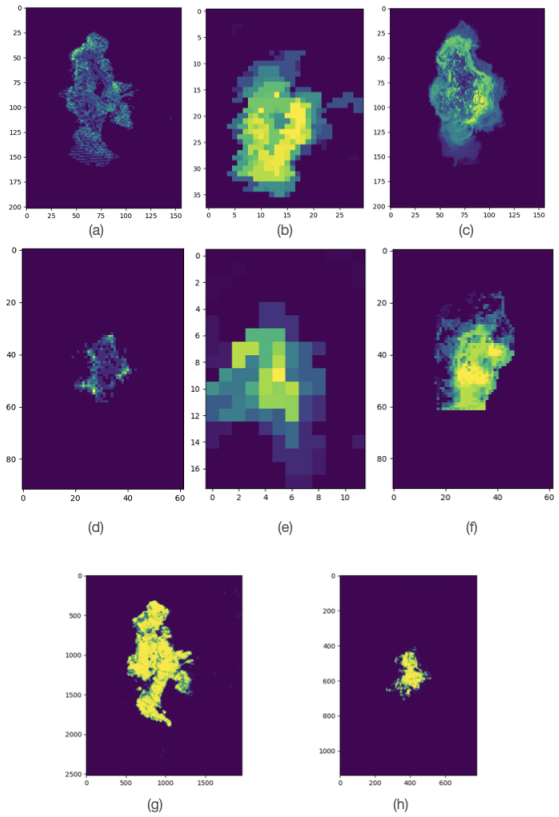


Fig. 4: Comparison for two different study areas, between (a) FIRMS The Creek Fire, (b) GOES-R fire standard product, The Creek Fire, (c) Deep GRU network, The Creek Fire, (d) FIRMS, Doctor Creek Fire, (e) GOES-R fire stadard product, Doctor Creek Fire, (f) Deep GRU network, Doctor Creek Fire, (g) Sentinel-2, The Creek Fire, (h) Sentinel-2, Doctor Creek Fire

$$NBR = \frac{Band_{11} - Band_{12}}{Band_{11} + Band_{12}} \quad (4)$$

The proposed method is also compared with FIRMS VIIRS fire product and GOES-R fire product in the two validation areas. The results are presented in Fig. 4 showing the improved resolution of the results obtained with Deep GRU (c,f) with respect to the standard GOES-R products (b,e). Moreover, the results reported a good visual agreement with the Sentinel-2 burned area reference maps.

Table 1: Earliest detection time comparison between FIRMS and Deep GRU network

Study Area	Deep GRU Network	FIRMS
Doctor Creek Fire	18:00-19:00	20:18
The Creek Fire	06:00-07:00	09:06

In Table 1, we compared the earliest fire detection time between FIRMS products and our results. The proposed method can detect the wildfire around two hours early than FIRMS in both validation areas, exploiting GOES-R satellites' high temporal resolution.

For burned area assessment, Sentinel-2 burned area maps were used as references. Considering that there is a substantial gap in terms of resolution between the Deep GRU results and the Sentinel-2 burned area maps, we adopted the following strategy to compute the standard evaluation metrics. i.e. Error of Commission and Error of Omission. We defined the criteria of true positive if the area covered by one low-resolution pixel (375m x 375m) contains at least one fire pixel (20m x 20m) in the higher resolution reference map. In this way, Error of Commission is defined as false alarms, where the image patch from Sentinel-2 corresponding to one fire pixel area at low-resolution does not contain any burned areas. Similarly, Error of Omission is defined where the image patches from Sentinel-2 identify small burned areas, but low-resolution results fail to detect. The assessment results are reported in Tab. 2 and 3.

Table 2: Burned area mapping Error of Commission for FIRMS, GOES-R, Deep GRU Network

Study Area	FIRMS	GOES-R	Deep GRU
Doctor Creek Fire	14.86%	75%	69.8%
The Creek Fire	13.24%	44.13%	36.66%

Table 3: Burned area mapping Error of Omission for FIRMS, GOES-R, Deep GRU Network

Study Area	FIRMS	GOES-R	Deep GRU
Doctor Creek Fire	20.07%	25.81%	5.78 %
The Creek Fire	22.11%	42.63%	7.11%

As shown in the table 2, compared with VIIRS active fire product, the proposed method have lower error of omission, but overestimates burned areas due to GOES-R's low spatial resolution. Compared with GOES-R active fire product, the results show that proposed method reduces both error of commission and error of omission. However, for Doctor Creek Fire in British Columbia, due to high slant angle of geostationary satellites in high latitude region, each pixel has much lower spatial resolution. Thus, the proposed method has higher error of commission but lower error of omission.

5. CONCLUSION

In this study, Deep GRU network is proposed to detect active fire pixels from GOES-R time series in the early stage of the wildfire. By applying resampling techniques and using 375 m spatial resolution remote sensing image as ground truth during the training, the proposed network can detect the wildfire earlier than the state-of-the-art fire products developed from VIIRS images. At the same time, this network also produce decent accuracy for burned area mapping and the result outperforms the fire product developed by NASA using GOES-R images.

6. REFERENCES

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